Junseo Kim

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Education

Seoul National University — Seoul, South Korea
B.S., Rural Systems Engineering — Mar. 2022 – 2028 (expected)

- Second Major: Artificial Intelligence (Interdisciplinary Program)
- GPA: 3.89 / 4.30 (as of Spring 2026)

Research Interests

Physical AI and Embodied Intelligence

Interested in multimodal learning, computer vision, and robot learning, particularly how AI systems perceive, reason about, and interact with the physical world.

Selected Projects

Insurance Charge Prediction | Machine Learning Fundamentals — Spring 2026

- Implemented LightGBM/CatBoost quantile-regression ensembles and a PyTorch MLP for tabular prediction under an asymmetric loss function.
- Analyzed distribution shift and validation leakage; applied feature engineering, cross-validation, and model blending (reported held-out penalized error: ~1,735).

SNU Lab Finder | Computer Programming — Spring 2026

- Developed a Java-based research lab recommendation system using object-oriented design and the Strategy Pattern to match student goals and preferences.
- Implemented recommendation explanations, lab comparisons, and persistent profiles based on 35 simulated research-lab records.

Streamflow Modeling with LSTM and SWAT | Engineering Project — Spring 2026

- Applied an Optuna-tuned LSTM to watershed streamflow prediction (R-squared = 0.71, NSE = 0.69) and calibrated a SWAT hydrological model.
- Compared data-driven and process-based simulations, analyzing prediction errors and limitations in the underlying data.

Technical Skills

Programming & Software	Python, Java, C++, JavaScript
Machine Learning	PyTorch, scikit-learn, LightGBM, CatBoost, Optuna
Computing & Modeling	Conda, SWAT
Languages	Korean (Native), English (Proficient)

Relevant Coursework

AI / ML	Machine Learning Fundamentals; Deep Learning Fundamentals (in progress, Fall 2026)
Computer Science	Computer Programming; Data Structures; Algorithms (in progress, Fall 2026)
Mathematics	Engineering Mathematics I & II; Applied Engineering Mathematics I & II

Honors and Awards

Champion & Finals MVP, 1st SNU Cup — 2026
National Collegiate Basketball Tournament

Activities & Service

Republic of Korea Army — Jul. 2024 – Jan. 2026
Completed mandatory military service (honorably discharged).

SNU Department of Physical Education Basketball Team — Mar. 2022 – Dec. 2024
Registered player, collegiate basketball Division II.

HOBAS, SNU Basketball Club — 2026 – Present
Member